

Backend Architecture Research

1. RESTful API Architecture

The advantages of REST architecture include

- Simplicity
- Stateless communication
- Web and mobile clients' compatibility

REST APIs are perfectly suited for educational platforms and learning management systems. (Fielding, 2000).

2. Modular Architecture

Modular separation was used in the backend's design:

- Module for authentication
- Module for profiles
- Module of exercises
- AI helper module
- Module for Analytics

This enhances future extensibility, testability, and maintainability.